# Design capacity expands. Validation must keep up.

Illustrative stage comparison, normalized per one clinical entrant. Supplied assumptions, not cohort statistics.

| STAGE | PRE-AI | AI ERA | WHAT CHANGES |
|---|---|---|---|
| Molecules designed (virtual) | ~$10^3$ | $10^5$–$10^9$ | Generative models can expand candidate enumeration at low marginal cost. |
| Molecules physically made | ~$10^3$ | $10^3$–$10^4$ | Automated synthesis and DNA libraries can expand construction capacity. |
| **Fully profiled in wet lab** | **~50** | **50 → 500** | **Illustrative capacity expansion that investors would need to fund.** |
| Clinical entry | 1 | 1 | Patients, ethics, capital and time remain hard constraints. |
| Expected approvals | ~0.14 | Unknown | Clinical success must be demonstrated; AI does not establish an approval rate. |

**Working thesis: experimental throughput becomes a capacity gate as virtual design scales.**

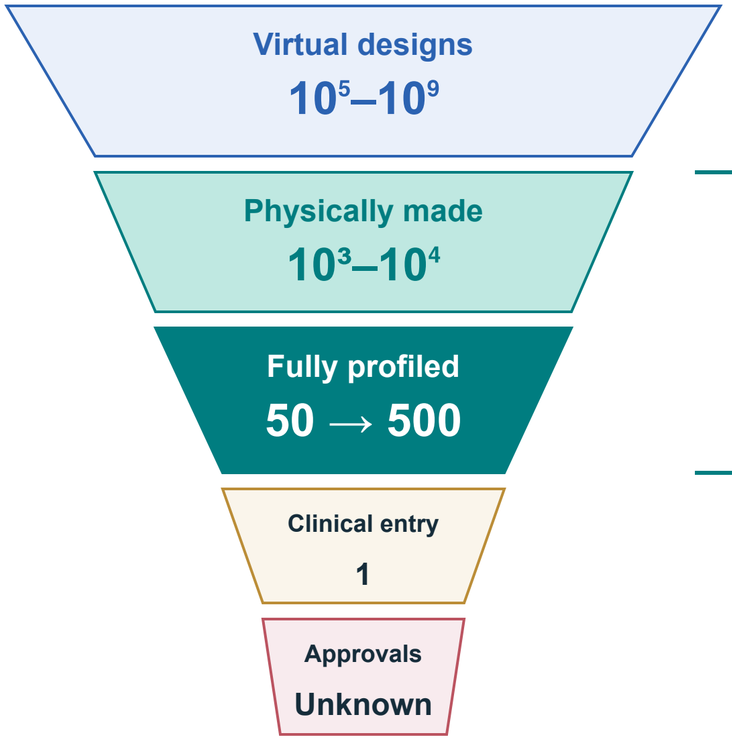
Source: user-supplied comparison. ~0.14 is an illustrative expected approval count per clinical entrant; the AI-era outcome is unknown.

# AI-scale design meets finite experimental capacity

Working scenario: candidate generation expands faster than physical construction and biological validation.

## AI-ERA DISCOVERY FUNNEL

Illustrative counts per one clinical entrant




**WET LAB = BUILD + TEST**

Funnel widths are schematic.
Clinical success remains unproven in this scenario.

## The LST ecosystem behind the capacity gate

Selected market anchors • different scopes and years • do not add these values

### Molecular construction

DNA / RNA synthesis, enzymes
Chemical building blocks

**RMB 1.33bn**

China • 2025 • DNA synthesis services [1]

Oligos + long fragments only

TWIST BIOSCIENCE · GenScript Make Research Easy

Twist • GenScript • Tsingke • Vazyme

### Protein & functional assays

Recombinant proteins, antibodies
Binding and activity assays

**US$3.5bn**

Global • 2025 • research proteins [2]

China: ~US$0.4bn • RUO scope

ACRO BIOSYSTEMS

ACROBiosystems • Bio-Techne • Sino

### Cells & culture systems

Cell models, media, feeds
Bioreactors and cell expansion

**US$323.5m***

China • 2024 • bioprocess media [3]

Selected media anchor; not all cell tools

**Gibco | OPM | Duoning**

Thermo / Gibco • OPM • Duoning

### Labware & sample handling

Tips, plates, tubes, filtration
Automation-compatible labware

**~RMB 23bn**

China • 2024 • plastic labware [4]

Includes clinical and industrial uses

BIOFIL

JET Biofil • Corning • Changhong

### Measurement & analytics

Sequencing, LC / LC-MS, flow
Imaging and quality control

**RMB 4.3bn**

China • 2024 • sequencing tools [5]

Sequencers + consumables; all uses

MGI

MGI • Illumina • Agilent • Waters

### Purification & scale-up

Analytical columns, process resins
Filtration and single-use systems

**US$578.2m***

China • 2024 • bioprocess resins [3]

Manufacturing demand has a longer lag

NanoMicro · cytiva

NanoMicro • Cytiva • Sepax

**Automation + data orchestration span every tool family** — TECAN

[1] Tsingke IPO 2026 [2] ACRO HKEX 2026 [3] Duoning HKEX 2026* [4] JET 2025 [5] MGI 2024
*Previously checked official-index data; source PDFs currently unavailable. Scope and citations: SOURCES.md.

# 01 The life science tools framework

FRAMEWORK

A product framework spanning academic research, biotech, pharma, CRDMOs and manufacturing.

## Reagents

| | |
|---|---|
| 01 | Recombinant proteins / antibodies |
| 02 | Enzymes / PCR and NGS reagents |
| 03 | DNA / RNA / chemical building blocks |
| 04 | Cell lines / culture media |
| 05 | Assay reagents for R&D and QC |

Tests × dose per test × price

## Consumables

| | |
|---|---|
| 01 | Tips / microplates |
| 02 | Culture flasks / centrifuge tubes |
| 03 | Membranes / functional beads |
| 04 | Columns / chromatography resins |
| 05 | Gels / process components |

Tests or batches × intensity × price

## Instruments

| | |
|---|---|
| 01 | Sequencers / liquid handlers |
| 02 | LC / LC-MS / mass spectrometers |
| 03 | Flow cytometers |
| 04 | High-content imaging systems |
| 05 | Cell-culture workstations |

New installations + replacement + service

Use one product taxonomy; retain the scope of each market estimate.

Classify products separately from their use cases. Chromatography resins can be reused.

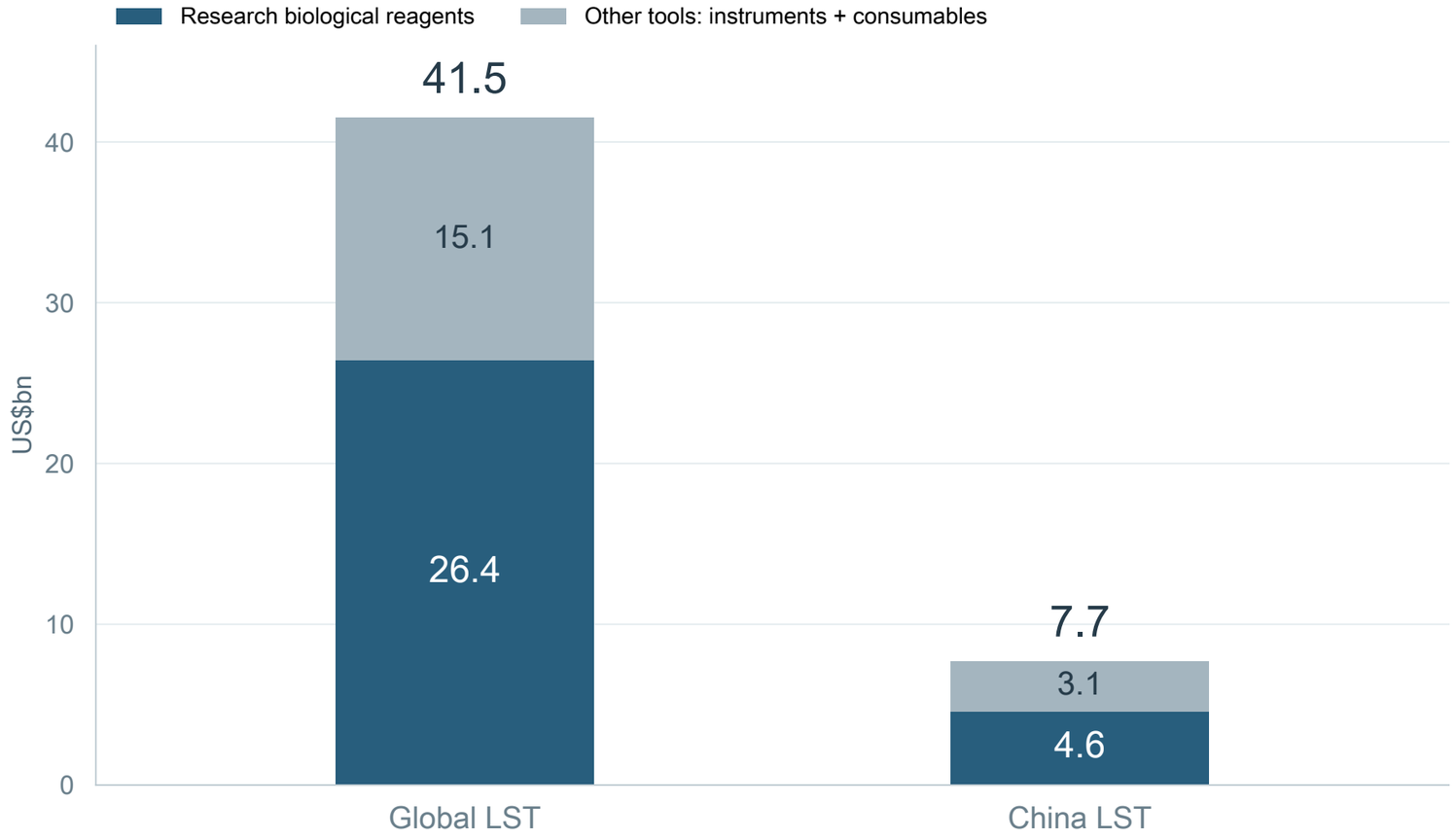
Source: supplied framework and prior product research. Existing TAMs do not automatically reconcile.

2026-10-12

# 02 China and global LST markets

HISTORICAL

2025 historical estimates from one F&S study; this is a defined market scope, not a universal LST TAM.




China / global market

18.6%

7.7 / 41.5
Same-scope historical estimate

Broader research and production markets require a separate scope reconciliation.

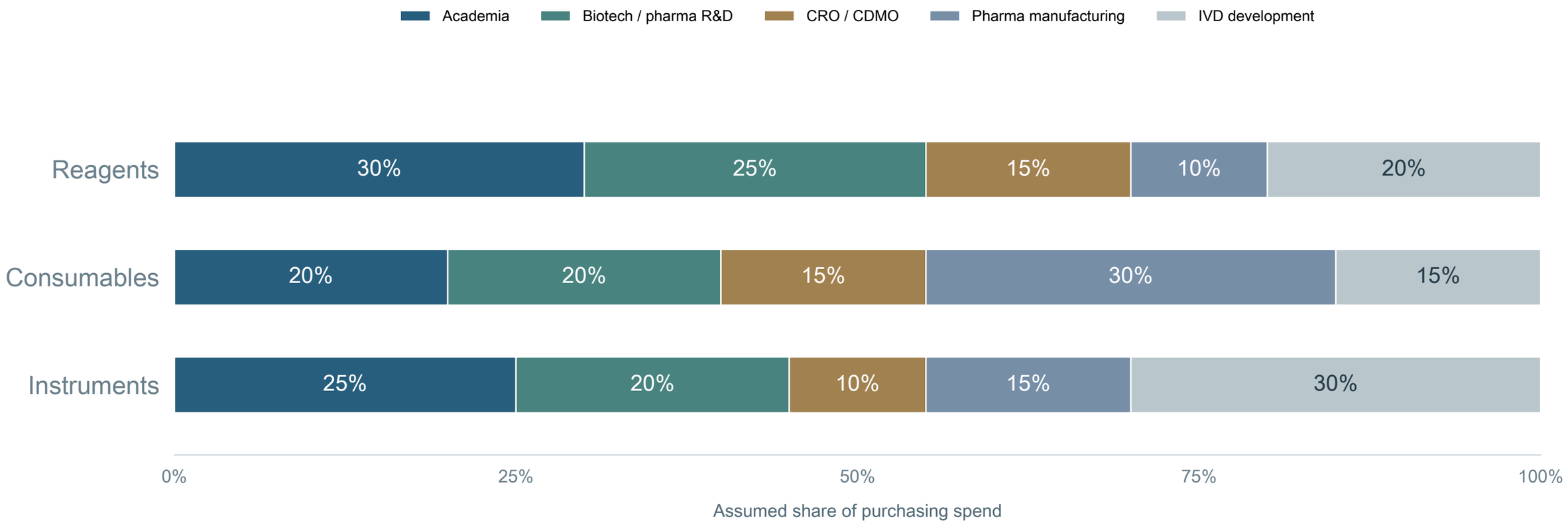
Other tools are the residual. Comparable instrument and consumable splits were not separately available.

Source: ACROBiosystems HKEX application, 30 Sep 2026, printed pp. 65 and 67; links in SOURCES.md.

# 03 How downstream demand differs

ILLUSTRATIVE

Direct purchasing differs by product basket. These supplied assumptions sum to 100% for each row.




Reagents: more research exposure in this scenario

Consumables: more production exposure

CRO/CDMO purchases are separate from pharma-owned production. Hospitals are not separately modeled.

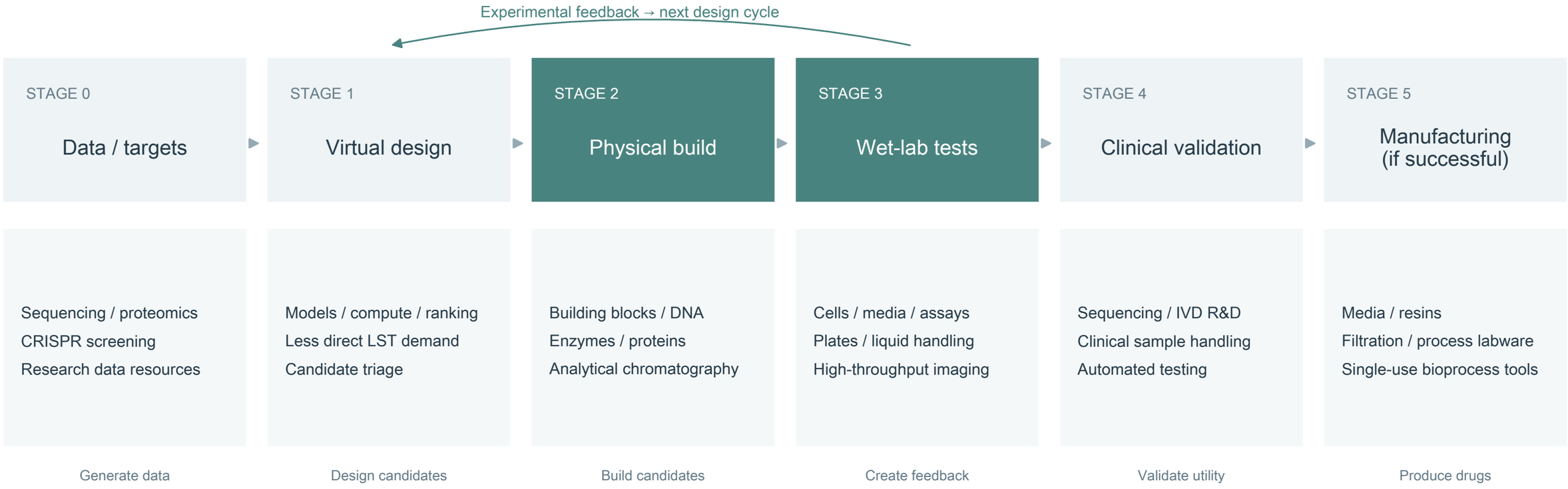
Source: illustrative shares in the supplied manuscript; these are not measured customer shares.

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# 04 AI discovery and its LST touchpoints

MECHANISM

AI candidates need physical construction and experimental feedback. Orders depend on funded projects.




The most direct potential uplift comes from building and testing candidates.

Stages 2–3 have more direct exposure. Clinical and manufacturing demand face attrition and time lags.

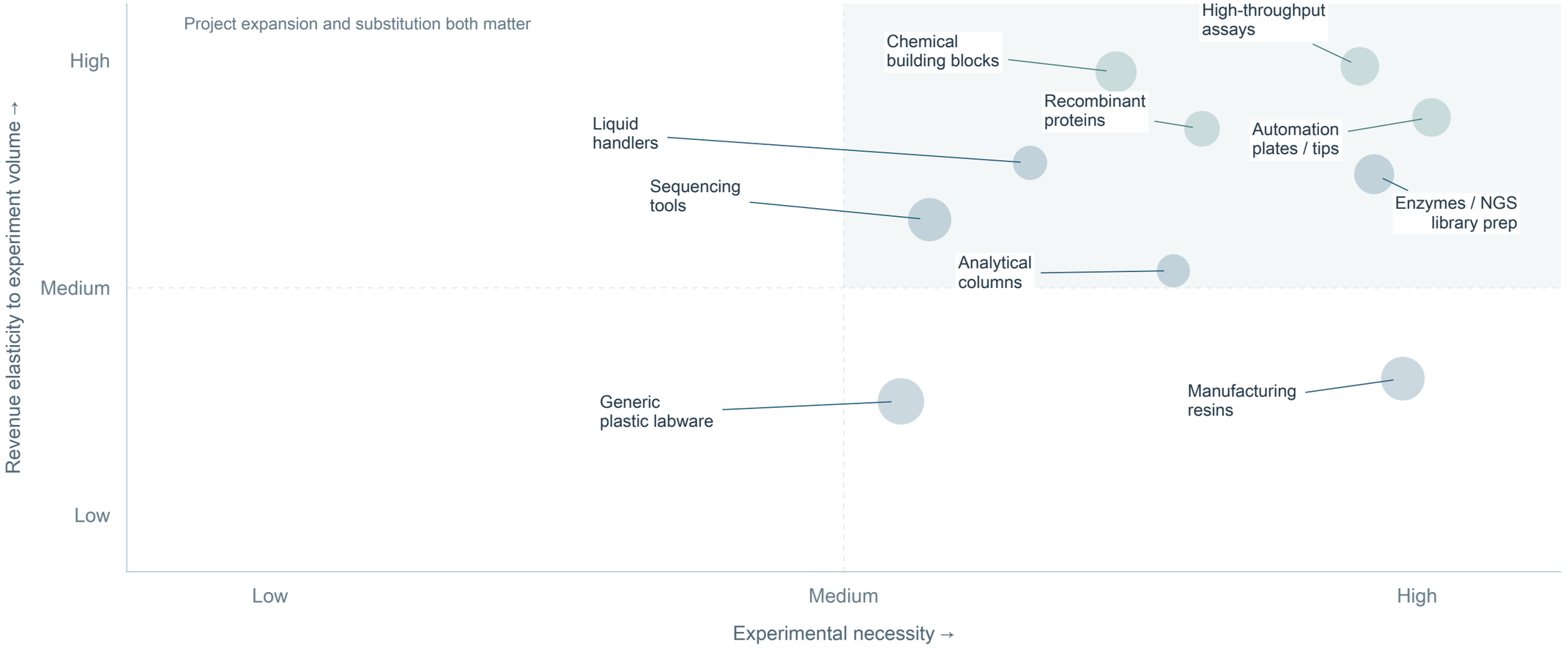
Source: supplied framework and product-use research. Arrows describe a mechanism, not realized orders.

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# 05 Which tools have more demand elasticity?

HYPOTHESIS

Positions and bubble areas are qualitative research judgments, not measured scores or additive TAMs.




Bubble area

Relative opportunity illustration; not measured TAM

Potential elasticity needs paid demand and repeat orders

Net demand also depends on consumption intensity, prices, project attrition and technical substitution.

Source: supplied framework and analyst assumptions. Edit positions and relative areas in data.json.

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# 06 More experiments, less material per experiment

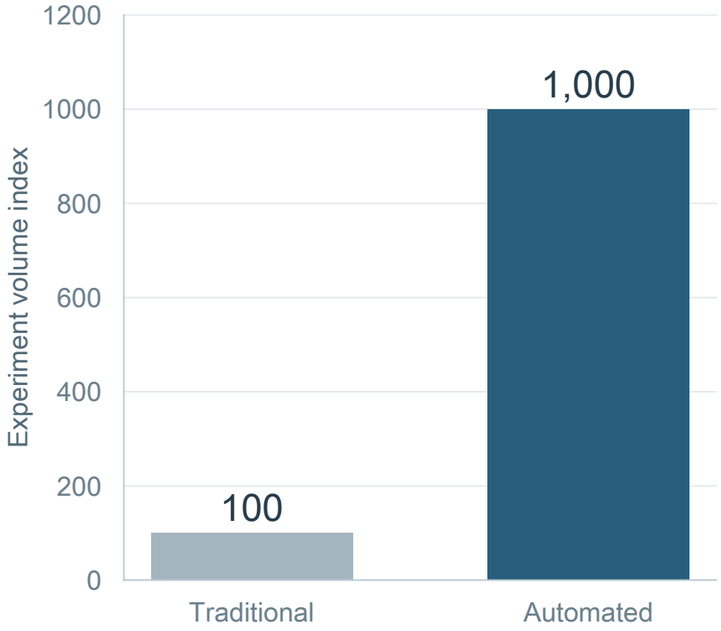
SCENARIO

Traditional: 100k tests, 96 wells, 100 µL/test. Automated: 1m tests, 1,536 wells, 5 µL/test. Base = 100.

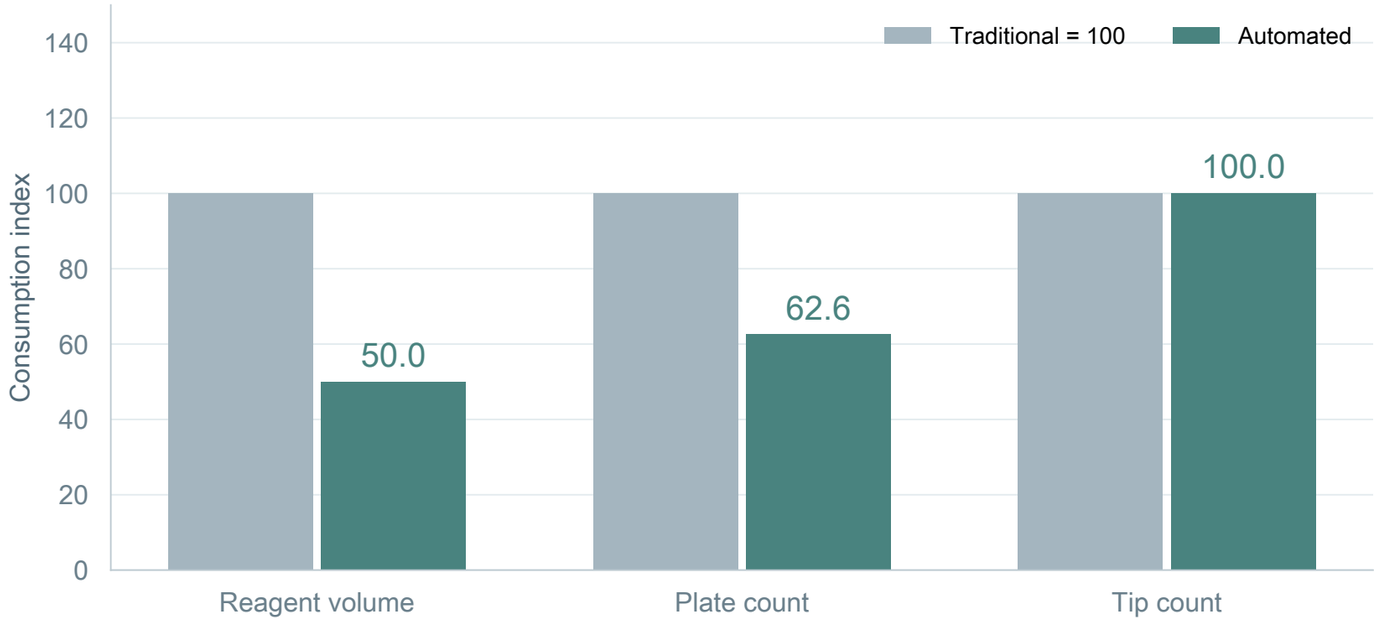
## 100k → 1m experiments

Reagents: 10 L → 5 L · Plates: 1,042 → 652 · Tips: 100k → 100k (assumed)

### Experiment volume: 10×




### Consumption can fall or remain flat




**Experiment growth does not imply proportional purchasing growth.**

Y-axis scales differ. Full plates, no dead volume or control wells; tips fall from 1 to 0.1 per test. ASP excluded.

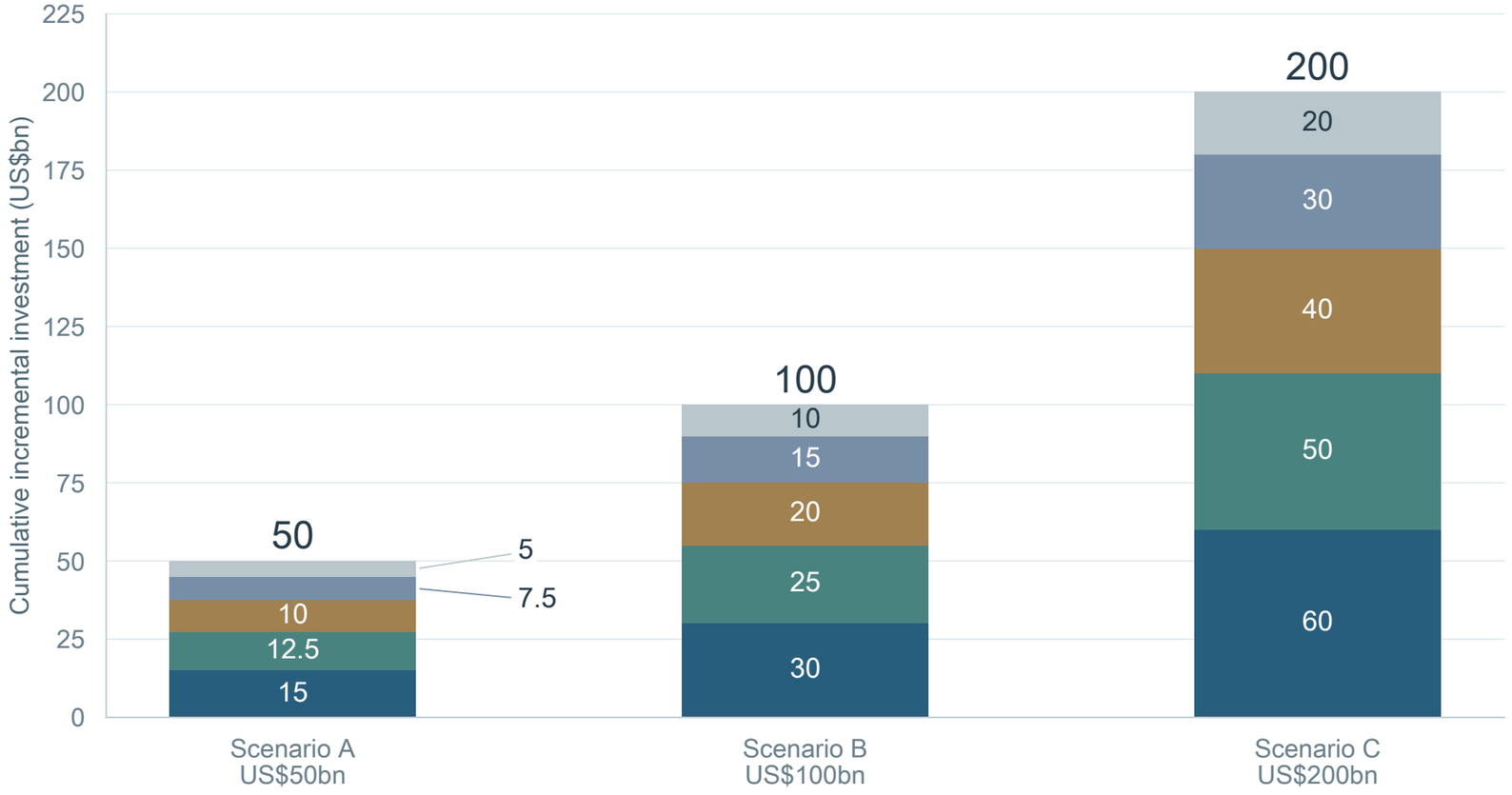
Source: supplied scenario and analyst calculation. Test definition and plate utilization affect the result.

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# 07 Illustrative AI-driven investment scenarios

SCENARIO

US$50bn / 100bn / 200bn are illustrative cumulative investment pools; no time horizon is specified.




Research equipment / automation
Research reagents
Research consumables
Manufacturing bioprocess tools
Services / other

Build phase → installed equipment

Operate → recurring purchases

Manufacturing needs drug output.

Allocations: 30% / 25% / 20% / 15% / 10%. Research and manufacturing buckets are mutually exclusive.

Source: supplied scenario and allocation assumptions. These are not realized funding, orders or annual TAMs.

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# 08 Potential supply constraints over 12–24 months

HYPOTHESIS

Relative expansion and qualification constraints over 12–24 months; current shortages are not established.




Track lead times, stockouts, qualification, yields and prices. Manufacturing resins have a longer demand lag.

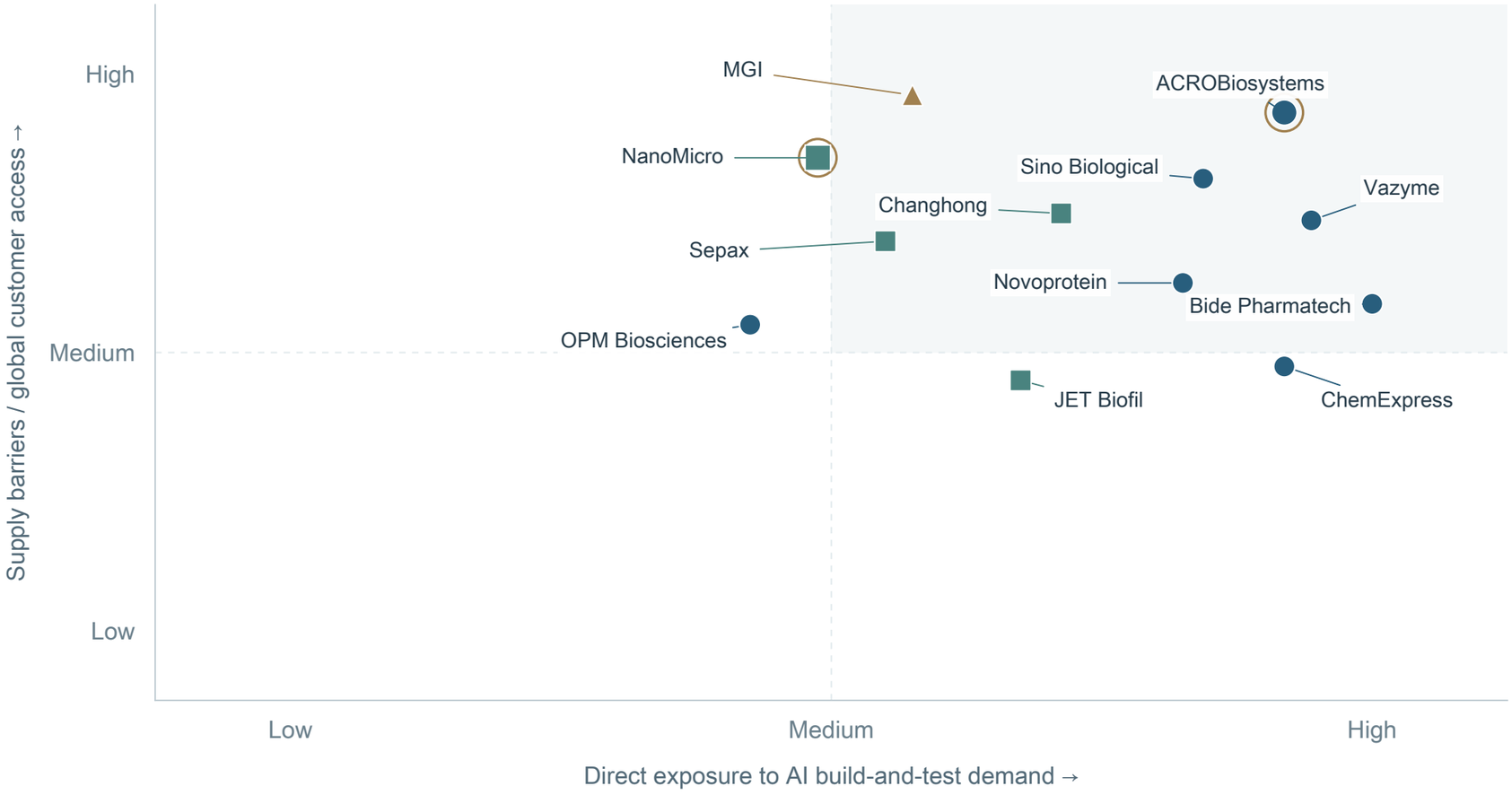
Source: supplied framework and analyst hypotheses. Technical barriers do not prove a capacity bottleneck.

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# 09 Mapping Chinese listed LST suppliers

HYPOTHESIS

Product exposure to AI build-and-test stages; positions do not measure AI revenue share or investment rank.




### Reagents

Proteins, enzymes, blocks, media

### Consumables

Labware, columns, resins

### Instruments / platforms

Sequencing and automation

**Two different demand pathways**

ACRO: expression / functional assays
NanoMicro: purification + scale-up

Axes are qualitative judgments. Qualification, global customer access and paid orders still require evidence.

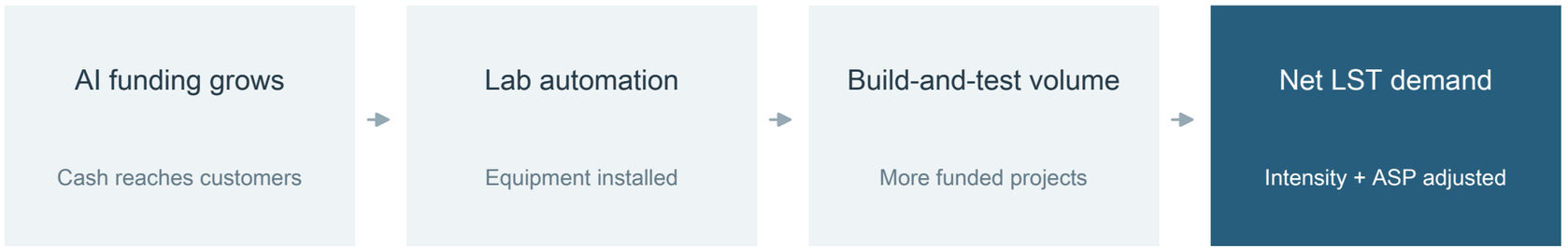
Sources: company filings and official product catalogs. Product rationale and assumptions are in data.json.

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# 10 From AI funding to LST shareholder returns

THESIS

Funding starts the chain. Projects, consumption, supply, market share and cash returns determine the outcome.




### Direct demand

Assays, automation labware and chemical building blocks

### Supply constraints

Specialty plates / components
Expansion limits need evidence

### Listed candidates

ACRO, Vazyme, Sino, NanoMicro JET, Changhong and others

### Evidence is needed at each step

Customer budgets → orders / repeat purchases → net consumption → utilization → company share → recurring profit / free cash flow → growth priced into the stock

Experimental relevance ∩ Evidenced supply constraints ∩ Global customer access

Seek experimental relevance, evidenced supply constraints and global access; test the valuation hurdle.

Source: supplied framework and prior company research. Capital announcements are not LST orders.

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